ERIC ZHAN

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EDUCATION

California Institute of Technology 2016 - 2021
Ph.D. Computing & Mathematical Sciences, GPA: 3.8
Thesis: New Algorithms for Programmatic Deep Learning with Applications to Behavior Modeling

Cornell University 2012 - 2016
B.A. Computer Science and Mathematics, *summa cum laude*, GPA: 4.0

RESEARCH & WORK EXPERIENCE

Argo AI - Senior Software Engineer Feb 2022 - Feb 2023

- Engineer in Autonomy group working on data-driven models for improving prediction and motion planning.
- Provided guidance and feedback on a wide range of machine learning projects across Autonomy, such as classifying agent behaviors, forecasting agent trajectories, and learning scene representations.
- Deployed a machine learning model to predict non-yielding probabilities of actors in a scene, allowing the autonomous vehicle (AV) to better avoid driving scenarios in which human takeovers are required. Responsibilities include: creating the dataset for training and testing, designing a deep model in PyTorch, preparing metrics for evaluation, and integrating the model into a multithreaded latency-critical environment on the AV.
- Designed a lightweight environment (dataset construction and serialization, dataclass schemas, simulation, evaluation metrics) to enable rapid prototyping and evaluation of black box motion planners.

Caltech - Machine Learning Researcher (Advisor: Yisong Yue) Sep 2016 - Nov 2021

- Conducted research in behavior modeling and sequential decision making, focusing on developing new methods for imitation learning inspired by generative models, weak supervision, and program learning.
- Thesis explores two themes: using domain-specific programs as sources of weak labels and using programs as a flexible function class with interpretable structure for neurosymbolic program learning.
- Published research papers at top conferences for machine learning (ICML, ICLR, NeurIPS, CVPR).

Facebook - Machine Learning Engineer Intern Summer 2021

- Explored several approaches for identifying credible threats of violence (CTV) by leveraging social graphs with millions of nodes (users, posts, groups) and edges (friendship, group membership).
- Trained graph neural networks using both supervised and unsupervised learning methods, the latter of which resulted in a 5x improvement in recall on the held-out "golden set" of CTV examples.

Argo AI - Research Intern Summer 2020

- Designed deep models and learning objectives to learn compact representations of complex traffic scenes, and leveraged learned representations to enable efficient search and retrieval of similar traffic scenes.
- Developed new tool for scene retrieval, improving dataset quality and feedback signals for model evaluation.

Microsoft - Research Intern Summer 2018

- Investigated reinforcement learning (RL) and imitation learning (IL) methods for calibrating agents to specific playstyles for playing Atari games.
- Implemented and benchmarked various state-of-the-art RL & IL algorithms in Project Athens codebase.

SKILLS

Programming: Python, SQL, C++, Java
Tools: PyTorch, TensorFlow, Keras, LaTex, Jupyter, Colab, EC2

PUBLICATIONS

Eric Zhan. New Algorithms for Programmatic Deep Learning with Applications to Behavior Modeling. Dissertation (Ph.D.), California Institute of Technology, 2022. https://thesis.library.caltech.edu/14436/.

Eric Zhan*, Jennifer Sun*, Ann Kennedy, Yisong Yue, Swarat Chaudhuri. Unsupervised Learning of Neurosymbolic Encoders. *TMLR 2022*.

Jennifer Sun, Ann Kennedy, **Eric Zhan**, Yisong Yue, Pietro Perona. Task Programming: Learning Data Efficient Behavior Representations. *CVPR 2021 (Best Student Paper Award)*.

Eric Zhan, Jagjeet Singh, Yisong Yue, Andrew Hartnett. The Argoverse Trajectory Retrieval Benchmark. *In submission.*

Oliver Stephenson, Tobias Köehne, **Eric Zhan**, Brent Cahill, Sang-Ho Yun, Zachary Ross, Mark Simons. Deep Learning-based Damage Mapping with InSAR Coherence Time Series. *IEEE Transactions on Geoscience and Remote Sensing 2021*.

Ameesh Shah*, **Eric Zhan***, Jennifer Sun, Abhinav Verma, Yisong Yue, Swarat Chaudhuri. Learning Differentiable Programs with Admissible Neural Heuristics. *NeurIPS 2020*.

Eric Zhan, Albert Tseng, Yisong Yue, Adith Swaminathan, Matthew Hausknecht. Learning Calibratable Policies using Programmatic Style-Consistency. *ICML 2020*.

Yukai Liu, Rose Yu, Stephan Zheng, **Eric Zhan**, Yisong Yue. NAOMI: Non-Autoregressive Multiresolution Sequence Imputation. *NeurIPS 2019*.

Eric Zhan, Stephan Zheng, Yisong Yue, Long Sha, Patrick Lucey. Generating Multi-Agent Trajectories using Programmatic Weak Supervision. *ICLR 2019*.

Eric Zhan, Stephan Zheng, Yisong Yue. MAGnet: Generating Long-Term Multi-Agent Trajectories. *Bayesian Deep Learning workshop, NeurIPS 2017*.

*equal contribution

ACADEMIC EXPERIENCE

Teaching Assistant 2016 - 2021

- Caltech: CS101 Projects in Machine Learning, CS159 Advanced Topics in Machine Learning.
- Cornell: CS4820 Analysis of Algorithms, CS4850 Mathematical Foundations for the Information Age.

Conference Reviewing

- NeurIPS 2020, ICLR 2021, CVPR 2021.

SOFTWARE ENGINEERING EXPERIENCE

LinkedIn - Software Engineering Intern Summer 2015

- Engaged in product decisions to improve the social connections that users can build through LinkedIn.
- Implemented new messaging features for LinkedIn's desktop and mobile web applications.

Yahoo - Software Engineering Intern Summer 2014

- Designed a database of fantasy sports user data to study factors that affect user retention rate.

HONORS & AWARDS

Cornell Computer Science Honors Program *summa cum laude*. 2016
Cornell Pauline and Irving Tanner Dean Scholar. 2012
Honorable Mention at 42nd International Physics Olympiad. 2011